SIMULATION AND ENGINEERING ANALYSIS OF A GAS–OIL SEPARATION PLANT (GOSP)

An Equipment-Centered Academic Report Using Aspen HYSYS

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Project Report

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ABSTRACT

This report develops and interprets an integrated steady-state simulation of an oil, gas and produced-water treatment facility using Aspen HYSYS V14. The modeled flowsheet includes high-, low- and low-low-pressure three-phase separators; heat integration; a flash-based dehydration representation; two modeled wash-water desalting stages; a 12-stage crude stabilizer; gas gathering and two main export compression stages; cooling and knockout drums; water handling; and export pumps. The objective is not only to reproduce a flowsheet but to explain the governing physical principles, define inlet and outlet streams for each unit, corrosion assessment modeling for streams with analyzing, show relevant calculations and critically evaluate the credibility and limitations of model outputs.

At its reference operating point, three primary inlet streams supply approximately 792.66 tonnes/h before addition of two wash-water feeds. The facility produces approximately 555.00 tonnes/h of export crude and 61.50 tonnes/h of export gas; the water-treatment route sends approximately 234.54 tonnes/h to the injection pump. The stabilizer uses 12 equilibrium stages and a feed at stage 6, with partial condensation, a 0.50 reflux ratio and a modeled bottoms temperature near 150°C. The exported energy stream results include 30.98 MW of reboiler heating and approximately 6.29 MW of total modeled rotating-shaft demand. Three discrete case studies evaluate throughput, reflux ratio and HPPT feed pressure. Their interpretation is based on measured outputs and model-boundary checks, not on unverified product-quality or operability assumptions.

1. INTRODUCTION AND ENGINEERING OBJECTIVES

1.1 Industrial context

The well stream delivered to a gas oil separation plant is rarely a single liquid. It normally contains a hydrocarbon gas phase, liquid hydrocarbons with dissolved light components, produced formation water, and potentially entrained water droplets and contaminants. Changes in pressure and temperature redistribute hydrocarbons between gas and liquid; water, because of its very different polarity and density, forms a separate aqueous phase when free water is present. Processing must therefore perform several connected tasks: separate phases, limit hydrocarbon loss to the water route, condition crude oil for downstream handling, recover gas at suitable pressure, and route aqueous streams for further treatment or injection.

These tasks cannot be interpreted as independent boxes. A separator pressure determines how much gas flashes from the oil; this affects gas-compressor throughput and the amount of light material entering the crude stabilizer. Desalting adds wash water and creates brine for the water-management circuit. Heating improves some separations but increases utility duty; the hot bottoms of the stabilizer can supply useful heat to cold crude. Engineering assessment is consequently an integrated mass, phase-equilibrium, hydraulic and energy problem.

1.2 Purpose and research questions

The report addresses four questions:

(i) What function does each process item perform and by what mechanism?
(ii) How can its mass and energy balances be verified against the Aspen HYSYS streams?
(iii) How do selected operating variables affect production rates, product volatility and utility loads?
(iv) Which quantities are simulated directly, which are estimated using external assumptions, and which require further experimental or vendor validation?

1.3 Scope

The validated scope is an integrated steady-state simulation with representative fluid characterization and three single-factor sensitivity studies. Each HYSYS result describes a calculated operating point rather than a time trajectory. Equipment geometry, retention time, valve travel, controller tuning, shutdown logic, transient surge, corrosion rate and regulatory compliance are not established merely by achieving HYSYS convergence.

2. THEORETICAL FUNDAMENTALS AND GOVERNING EQUATIONS

2.1 Conservation of total mass and species

At steady state, process inventories are invariant with time. The general balance for any conserved component is accumulation = inlet − outlet + generation − consumption. Since chemical reaction is not represented in the present separator/conditioning train, the generation and consumption terms are zero for each hydrocarbon or water component. This yields the primary correctness test of the simulation:

$$\frac{dM}{dt} = \Sigma\, \dot{m}_{in} - \Sigma\, \dot{m}_{out} = 0 \qquad (2.1)$$

$$0 = \Sigma\left(\dot{n}_{in} z_{i_{in}}\right) - \Sigma\left(\dot{n}_{out} y_{i_{out}}\right) \quad \text{for each component i} \qquad (2.2)$$

Here $\dot{m}$ is mass flow rate (kg/h), $\dot{n}$ is molar flow (kmol/h), and z and y denote appropriate component fractions. Component balances are necessary when two equipment outlet streams contain different hydrocarbon compositions even if their total mass balance closes. Internally recycled liquids must not be double counted in the overall facility balance.

2.2 The first law for process equipment

For a steady-flow control volume, shaft work, energy carried by material streams and external heat transfer must balance. Neglecting changes in velocity, elevation and chemical reaction enthalpy is usually justified for the

simplified process models, although a real pump or high-velocity compressor may require these terms for detailed design.

$$\dot{Q} - \dot{W}_s + \Sigma\, \dot{n}_{in} h_{in} - \Sigma\, \dot{n}_{out} h_{out} = 0 \qquad (2.3)$$

HYSYS reports many power and utility streams in kJ/h. The conversion is 1 kW = 3,600 kJ/h. The sign convention must be checked before a duty is reported: an energy stream representing heat rejected from a process unit is a cooling requirement even if the software displays it with the opposite sign to heating input.

$$Power\ (kW) = \frac{\left|Energy\ flow\ \left(\frac{kJ}{h}\right)\right|}{3600} \qquad (2.4)$$

2.3 Phase behavior, fugacity and flash separation

Oilfield fluids exhibit nonideal phase behavior. In a vapor–liquid equilibrium calculation, a hydrocarbon partitions between phases until its fugacity is equal in both phases. HYSYS uses a cubic equation-of-state package, Peng–Robinson, to estimate equilibrium and enthalpy. The presence of heavy crude pseudo-components requires representative fluid characterization; otherwise, calculated volatility, condensate yields and Reid vapor pressure may not reflect a measured crude assay.

$$f_i,V(T,P,y) = f_i,L(T,P,x) \quad \text{and} \quad K_i = \frac{y_i}{x_i} \qquad (2.5)$$

For an idealized two-phase flash of feed composition z_i and vapor fraction β, component balances imply $x_i = \frac{z_i}{[1 + \beta(K_i - 1)]}$. The vapor fraction is obtained by simultaneously satisfying the Rachford–Rice relation. A three-phase separator extends this equilibrium/phase-split concept to include an aqueous phase.

$$\frac{\Sigma z_i(K_i - 1)}{[1 + \beta(K_i - 1)]} = 0 \qquad (2.6)$$

Reducing pressure normally makes the dissolved lighter hydrocarbons less soluble in the liquid and releases a part to gas. Temperature and feed composition can change the direction and magnitude of individual-component recovery; therefore, a separator should not be judged by pressure alone.

2.4 Physical three-phase separation and residence time

Phase equilibrium predicts composition, but actual vessels also require sufficient holdup, disengagement area and mist control. Hydrocarbon gas rises under buoyancy, free water settles below oil, and the gas stream exits after entrained droplets have had an opportunity to separate. Industrial separators use inlets that reduce momentum, a gravity section, liquid level control, interface level control and mist-elimination internals, even where the HYSYS equilibrium separator symbol abstracts these features.

$$t_{res} = \frac{V_{working}}{Q_{liquid}} \qquad (2.7)$$

$$vg, allow \approx KSB \sqrt{\left[\frac{\rho L - \rho G}{\rho G}\right]} \qquad (2.8)$$

Equation (2.7) defines an approximate retention time, while Equation (2.8) is the Souders–Brown form for superficial vapor-capacity screening. KSB is an empirically selected capacity parameter and must not be assigned without vessel service and internals information. Neither expression by itself establishes the adequacy of the actual HPPT, LPPT or LPDT design.

2.5 Dehydration, coalescence, and crude desalting

Free water can be removed from crude by residence time and enhanced droplet coalescence. An industrial electrostatic dehydrator may use electric fields to increase droplet collision/coalescence, reducing water carryover in crude. Salt removal is a different requirement: dissolved chlorides are concentrated in the water phase and may remain trapped as microscopic salty droplets within crude. Desalting therefore mixes comparatively clean wash water with crude so that salts transfer to the water, followed by separation of the

aqueous brine. Temperature, mixing intensity, wash-water quantity, droplet size and residence time all influence performance.

$$\eta_{salt} = \left[1 - \frac{S_{out}\dot{m}_{oil_{out}}}{S_{in}\dot{m}_{oil_{in}}}\right] \times 100\% \quad (2.9)$$

In Equation (2.9), S represents salt mass per unit crude mass on a consistent basis. This efficiency cannot be reported numerically for the current model, which contains a water wash and phase separation representation but no validated inlet/outlet salt measurements or explicit salt-removal physics. The modeled brine rate is not itself proof of desalting effectiveness.

2.6 Crude stabilization and multistage contact

A crude stabilizer strips lighter, more volatile components from oil to reduce volatility prior to storage or transportation. Gas and liquid flow countercurrently through contact stages. Heat supplied to the reboiler vaporizes light components at the bottom, while condensation at the top returns part of the condensed overhead as reflux. Because reflux and reboiler heat alter internal liquid and vapor circulation, optimization must consider both product quality and utility demand.

$$R = \frac{L_{reflux}}{D_{liquid}} \quad (2.10)$$

$$\text{At each stage j: } \Sigma\, \dot{n}_in_j - \Sigma\, \dot{n}_out_j = 0;\ y_{ij} = K_{ij}x_{ij} \quad (2.11)$$

The stage calculations are often described by MESH equations: material balances, phase-equilibrium relations, summation of component fractions and heat balances. A model with 12 theoretical stages does not automatically guarantee 12 actual trays; actual tray count depends on efficiency and internals design. Reid vapor pressure is a standardized empirical product property, distinct from true vapor pressure. The reported HYSYS Reid VP should be validated against an appropriate assay or test before claiming sales-specification compliance.

2.7 Compression, pumps and cooling

Compression raises the pressure of separated gas to the next downstream header or export requirement. Gas temperature normally rises during compression, motivating intercooling and liquid removal before subsequent compressor stages. The approximate ideal-gas isentropic work relation illustrates sensitivity to pressure ratio, absolute inlet temperature and heat-capacity ratio; HYSYS applies a real-fluid performance model using the specified efficiency.

$$w_{ideal} \approx \left[\frac{k}{k-1}\right]\left(\frac{R_{gas}T_{in}}{MW}\right)\left[\left(\frac{P_{out}}{P_{in}}\right)^{\frac{k-1}{k}} - 1\right] \quad (2.12)$$

$$\dot{W}_{actual} \approx \dot{m}\, w_ideal/\eta_s \quad \text{(illustrative isentropic form)} \quad (2.13)$$

Liquid pumps, by contrast, can be approximated using the pressure rise and volumetric flow, adjusted for hydraulic efficiency. The difference between a compressor drive load and a condenser thermal duty is critical to energy economics.

$$\dot{W}_{pump} \approx Q_{vol}\frac{\Delta P}{\eta_{pump}} \quad (consistent\ SI\ units) \quad (2.14)$$

$$Qcool \approx \dot{m}\, Cp\, (T_{in} - T_{out}) \quad (single-phase\ approximation) \quad (2.15)$$

A multi-phase cooler may condense hydrocarbons and release latent heat, so the correct exchanger duty should be computed from inlet and outlet enthalpy rather than only a constant Cp approximation. Heat recovery through a crude-to-crude exchanger reduces external utilities when approach temperatures and exchanger fouling constraints remain acceptable.

2.8 Steady-state sensitivity and its interpretation

A sensitivity study changes one independent input or a deliberately linked group of inputs while holding other specifications consistent. It answers how converged steady-state outputs respond; it does not measure transient stability or dynamic controllability. A case study is meaningful only if the intended independent variable actually

changes and calculated outputs vary consistently. HYSYS “Failed Cases = 0” indicates numerical convergence of the trial states, not verification against plant data.

3. SIMULATION BASIS AND REFERENCE CONDITIONS

3.1 Software and thermodynamics

The calculation uses Aspen HYSYS V14 with a Peng–Robinson fluid package. Arabian Light Crude is represented through a set of pseudo-components combined with explicitly represented light hydrocarbons, acid gas and water. The original petroleum-assay linkage was detached during attempted dynamic conversion; compositional fractions were retained as entered. Final interpretations are restricted to the steady-state results contained in the supplied workbook, screenshots and calculated variables.

3.2 Principal feed streams

Stream	T (°C)	P (kPa)	Mass flow (kg/h)
Arabian Light Crude	40.0	653	566,644
ASSOCIATED GAS	40.0	653	57,937
PRODUCED WATER	40.0	653	168,077

The mixed well fluid totals approximately 792,658 kg/h before two separate wash-water injections of about 33.6 tonnes/h each. Nominal pressures decrease through the three separator stages: HPPT ≈ 446 kPa, LPPT ≈ 239 kPa and LPDT ≈ 170 kPa. The desalter feed pressure is approximately 150 kPa; the column operates between about 120 and 140 kPa; export gas reaches approximately 2,980 kPa after compression and cooling.

3.3 Main modeling assumptions

The analysis treats the fluid as thermodynamically equilibrated at each HYSYS flash separator, without verified droplet-settling/entrainment efficiency. All principal reported cases are steady state. Pipe mechanical ratings, actual heat-transfer area, tower hydraulic flooding margin, salt chemistry, oil-in-water limit and equipment vendor curves are not supplied. Pump and compressor power values are read from the energy streams rather than guessed from mechanical efficiency equations. The estimated economic inputs are illustrative and do not represent a facility electricity or fuel contract.

4. INTEGRATED PROCESS FLOW AND SYSTEM BOUNDARY

The main process begins with Arabian Light crude, associated gas and produced water entering a well-fluid manifold. Following high-pressure separation, the oil passes through two lower-pressure separation stages and then through heating, dehydration and wash-water desalting. The conditioned crude enters the stabilizer, while the hot bottoms are heat-recovered, cooled and pumped to the export line. Liberated gases join the gas-gathering and compression network. Produced water and desalter brines enter water processing, with the model currently showing no separated recovered oil from WOSEP.

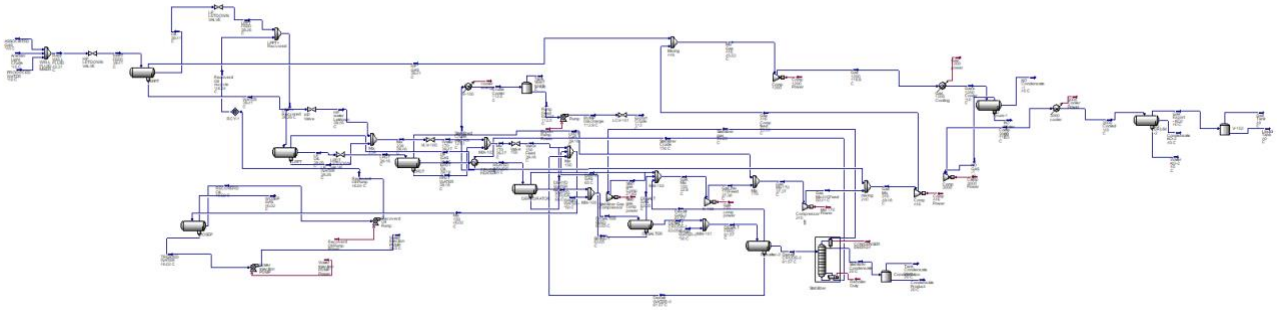


Figure 4.1. Integrated steady-state GOSP flowsheet exported from Aspen HYSYS V14.

For readability, equipment details and their actual stream names are developed individually in Chapter 5. Stream values refer to the model snapshot provided in the "Material Streams(4).xlsx" workbook, and not to illustrative industrial design benchmarks.

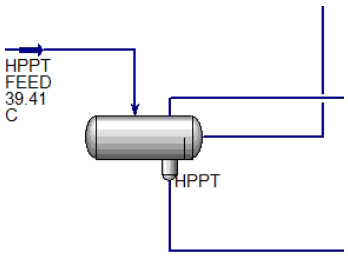
5. EQUIPMENT DESCRIPTION AND ENGINEERING ANALYSIS

This chapter follows the physical process rather than HYSYS interface order. For every item, the discussion gives its function, working principle, identified feeds and products, a local mass/energy or hydraulic relation, and the limitations of the simplified mathematical model. Small figures are supplied by HYSYS screenshots; a blank figure space intentionally indicates that an equipment screenshot has not yet been supplied.

5.1 High-Pressure Production Trap (HPPT)

Function. Primary three-phase separation immediately downstream of the combined well-fluid feed; it removes a large portion of free gas and bulk produced water before oil depressurization.

Working principle. The inlet momentum is reduced; gas disengages upward and gravity separates aqueous and hydrocarbon liquids. The equilibrium separator reports phase splits rather than mechanically calculating droplet sizes or mist carryover.



HPPT-supplied HYSYS image

Required input / reference	Simulated stream(s) and output(s)
Feed(s)	HPPT FEED: 792,658 kg/h.
Products / connected outlet(s)	HP GAS: 56,566; OIL: 568,700; WATER: 167,393 kg/h.
Representative operating conditions	446.0 kPa; 39.4 °C

$$\dot{m}_{HPPT_{feed}} \approx \dot{m}_{HP_{gas}} + \dot{m}_{oil} + \dot{m}_{water} \qquad (5.1)$$

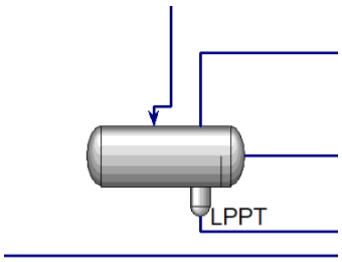
HPPTbp gas production is approximately 56.57 t/h at the nominal 446 kPa operating condition.

Engineering caution. Actual gas carry-under, oil-in-water and required vessel holdup must be checked separately from the thermodynamic flash.

5.2 Low-Pressure Production Trap (LPPT)

Function. Second-stage flash separator; it recovers additional gas liberated as the hydrocarbon liquid pressure falls following the HP letdown.

Working principle. Pressure reduction shifts light hydrocarbons toward vapor equilibrium. The LPPT removes newly released gas and sends the lower-pressure liquid toward LPDT; its modeled water outlet is zero for the current feed state.



LPPT -supplied HYSYS image

Required input / reference	Simulated stream(s) and output(s)
Feed(s)	LPPT FEED: 568,700 kg/h after the upstream HP letdown.
Products / connected outlet(s)	LP GAS: 935.1; LP OIL: 567,764; LP WATER: 0 kg/h.
Representative operating conditions	239.0 kPa; 39.3 °C

$$\dot{m}_{LPPT_{feed}} \approx \dot{m}_{LP_{gas}} + \dot{m}_{LP_{oil}} + \dot{m}_{LP_{water}} \qquad (5.2)$$

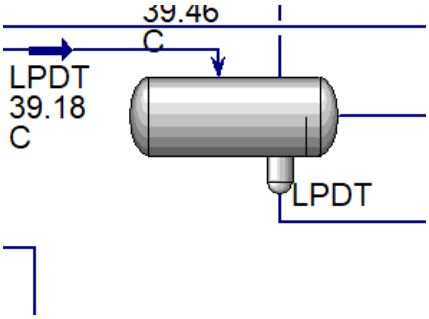
Only about 0.94 t/h additional gas is separated here under the model conditions.

Engineering caution. The zero LP-water result is a model outcome, not a claim that a field separator never accumulates water.

5.3 Low-Low-Pressure Production Trap (LPDT)

Function. Third-stage low-pressure degassing of the crude stream prior to heat treatment.

Working principle. Further pressure reduction causes residual dissolved light hydrocarbons to flash. The LPDT gas is routed into the gas gathering system, while its oil stream proceeds to crude conditioning.



LPDT -supplied HYSYS image

Required input / reference	Simulated stream(s) and output(s)
Feed(s)	LPDT: 567,764 kg/h.
Products / connected outlet(s)	LPDT GAS: 424.9; LPDT OIL: 567,339; LPDT WATER: 0 kg/h.
Representative operating conditions	170.0 kPa; 39.2 °C

$$\dot{m}_{LPDT\,feed} \approx \dot{m}_{LPDT\,gas} + \dot{m}_{LPDT\,oil} + \dot{m}_{LPDT\,water} \qquad (5.3)$$

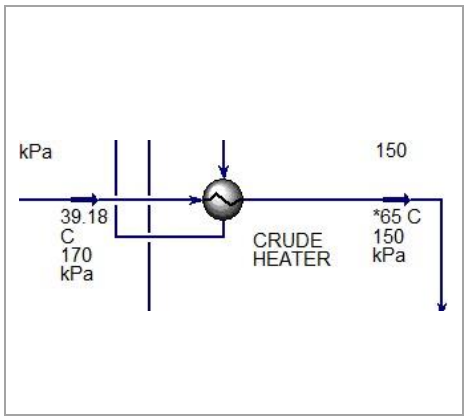
Pressure reduction from LPPT toward the 170 kPa LPDT level releases approximately 425 kg/h of gas.

Engineering caution. Gas flow may be sensitive to hydrocarbon pseudo-component characterization and interstage temperature changes.

5.4 Crude-to-Crude Heat Exchanger / Crude Heater

Function. Raise the temperature of low-pressure crude before dehydration and desalting while recovering energy from hot stabilized oil.

Working principle. Hot stabilized bottoms transfer sensible heat through the exchanger wall to colder inlet crude; the hot stream loses enthalpy while cold feed gains enthalpy. Heat exchange can also alter flashing and gas release. A process heater supplies any additional heat required by the crude conditioning system.



Required input / reference	Simulated stream(s) and output(s)
Feed(s)	Cold side: LPDT OIL (567,339 kg/h); hot side: Stabilizer Crude (554,995 kg/h).
Products / connected outlet(s)	HEATED CRUDE: 567,339 kg/h at 65°C; Stabilized Crude after HX: 127.1°C.
Representative operating conditions	Cold crude ~39.2 to 65°C; stabilized bottoms ~150 to 127.1°C (stream results).

$$\dot{Q}_{hot} = \dot{m}_{hot}(h_{in} - h_{out}) \approx \dot{Q}_{cold} + \dot{Q}_{loss} \qquad (5.4)$$

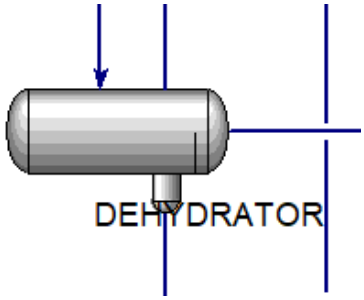
The equation represents first-law exchange; detailed heat-transfer design would additionally use $\dot{Q} = U\,A\,\Delta T_{lm}$.

Engineering caution. Exchanger mechanical area, fouling factors and pinch temperature are not validated by the snapshot.

5.5 Dehydrator

Function. Reduce separable free water associated with crude before the desalting train.

Working principle. In an actual facility, residence time and coalescence help remove water droplets; electrostatic coalescers use an electrical field where installed. In the supplied HYSYS model, the unit operates as a phase separator after crude heating and reports a small liberated gas stream.



DEHYDRATOR -supplied HYSYS image

Required input / reference	Simulated stream(s) and output(s)
Feed(s)	HEATED CRUDE: 567,339 kg/h.
Products / connected outlet(s)	DEHYD GAS: 1,159.1; DEHYD CRUDE: 566,180; DEHYD WATER: 0 kg/h.
Representative operating conditions	150.0 kPa; 65.0 °C

$$\dot{m}_{heated\ crude} = \dot{m}_{dehyd_{crude}} + \dot{m}_{dehyd_{gas}} + \dot{m}_{dehyd_{water}} \qquad (5.5)$$

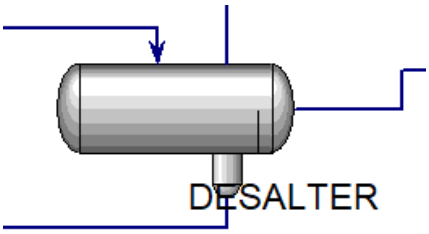
About 1.16 t/h gas flashes from the conditioned crude at 65°C and 150 kPa.

Engineering caution. Modeled zero aqueous draw does not establish real dehydration efficiency, which requires measurements of crude water content.

5.6 Desalter 1

Function. First wash-water-contacting and aqueous-phase separation stage for reducing potentially entrained water-soluble salts.

Working principle. Injected wash water is mixed into the crude to contact saline droplets; an actual desalter uses droplet coalescence and settling to reject a brine phase. The modeled separator returns a crude phase, brine and a small light-gas outlet.



DESALTER -supplied HYSYS image

Required input / reference	Simulated stream(s) and output(s)
Feed(s)	DEHYD CRUDE: 566,180 kg/h; WASH WATER-: 33,615 kg/h.
Products / connected outlet(s)	DESALT CRUDE-1: 565,996; BRINE-1: 33,512; DESALT GAS: 288 kg/h.
Representative operating conditions	DESALTER FEED: 599,796 kg/h; 150.0 kPa; 63.1 °C

$$\dot{m}_{crude_{in}} + \dot{m}_{wash1} \approx \dot{m}_{crude_{out}} + \dot{m}_{brine1} + \dot{m}_{gas1} \qquad (5.6)$$

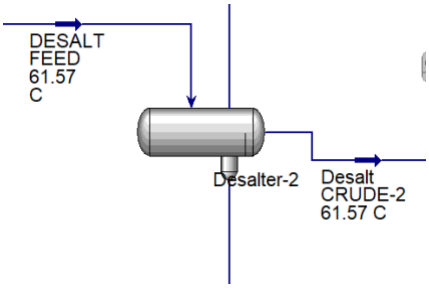
Wash water is approximately 33.6 t/h; the HYSYS brine draw is approximately 33.5 t/h.

Engineering caution. No validated NaCl species transfer, salt-in-crude ppm results, electrostatic-field model or desalting efficiency is claimed.

5.7 Desalter 2

Function. Second wash-water-contacting separation stage downstream of the first desalter.

Working principle. The second wash-water dose contacts residual soluble contamination and is subsequently separated as an aqueous outlet. In a real two-stage arrangement, washing and droplet coalescence can be operated countercurrently, depending on piping; the present flowsheet must not be assumed countercurrent without corresponding stream connections.



Desalter-2 -supplied HYSYS image

Required input / reference	Simulated stream(s) and output(s)
Feed(s)	DESALT CRUDE-1: 565,996 kg/h; WASH WATER-2: 33,621 kg/h.
Products / connected outlet(s)	Desalt CRUDE-2: 565,984; Desalt WATER-2: 33,633; Desalt GAS-2: 0 kg/h.
Representative operating conditions	DESALT FEED: 599,617 kg/h; 150.0 kPa; 61.6 °C

$$\dot{m}_{desalt_{feed}} \approx \dot{m}_{desalt_{crude2}} + \dot{m}_{desalt_{water2}} + \dot{m}_{desalt_{gas2}} \qquad (5.7)$$

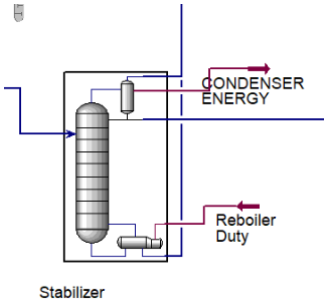
The processed crude leaves at about 565.98 t/h and 61.57°C.

Engineering caution. Without salt-bearing assays, two phase-separation stages cannot be interpreted as a quantified salt-reduction guarantee.

5.8 Crude Stabilizer (12 Stages)

Function. Remove light hydrocarbons from desalted crude and reduce the volatility of stabilized oil, while recovering overhead gas and condensate.

Working principle. The bottom reboiler produces upward vapor flow, which contacts descending liquid on equilibrium stages. A partial condenser produces an overhead gas and a condensate product, with 50% reflux relative to liquid distillate at the reference reflux-ratio specification. HYSYS uses 12 stages with feed introduced on stage 6.



Stabilizer -supplied HYSYS image

Required input / reference	Simulated stream(s) and output(s)
Feed(s)	Desalt CRUDE-2: 565,984 kg/h at 150.0 kPa; 61.6 °C.
Products / connected outlet(s)	Stabilizer Crude: 554,995; Stabilizer gas: 10,189; Stabilizer Condensate: 760.5 kg/h; condenser aqueous ≈39.2 kg/h.
Representative operating conditions	Partial condenser: 20°C and 120 kPa; reboiler: 150°C and 140 kPa; reflux ratio 0.50.

$$\dot{m}_{feed} \approx \dot{m}_{bottoms} + \dot{m}_{overhead_{gas}} + \dot{m}_{condensate} + \dot{m}_{aqueous} \qquad (5.8)$$

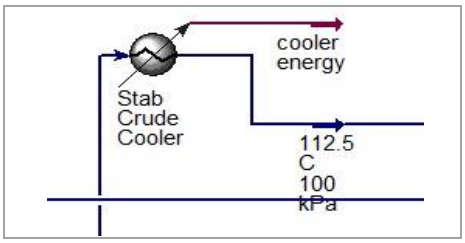
Reboiler Q = 111.517 × 10^6 kJ/h (30.977 MW); condenser heat-transfer stream magnitude = 3.074 × 10^6 kJ/h (0.854 MW).

Engineering caution. The 6.252 kPa model Reid VP does not by itself demonstrate ASTM-tested crude volatility or a contractual stabilization specification.

5.9 Stabilized Crude Cooler

Function. Reject remaining sensible heat from stabilized crude before export storage and pumping.

Working principle. Heat is removed by a cooling utility after initial recovery through the crude-to-crude heat exchanger. The stream is cooled without a deliberate material separation step.



Required input / reference	Simulated stream(s) and output(s)
Feed(s)	Stabilized Crude after HX: 554,995 kg/h.
Products / connected outlet(s)	Crude Cooler stream: 554,995 kg/h; T falls to 112.5°C.
Representative operating conditions	127.1°C to 112.5°C; energy stream "cooler energy": 5.00 MW.

$$\dot{Q}_{cool} = \dot{n}\,(h_{in} - h_{out}) \quad [enthalpy - based\ calculation] \quad (5.9)$$

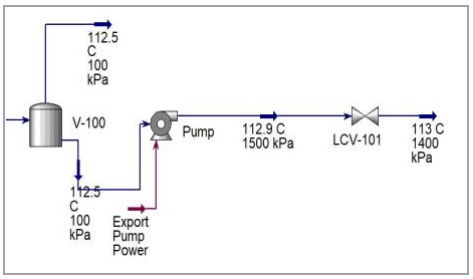
Use the 18.0×10⁶ kJ/h energy-stream duty, not the large signed material-stream "Heat Flow" used in the feed-rate case study.

Engineering caution. Actual cooling-media selection and approach temperature are not represented by the material stream alone.

5.10 Export Tank and Crude Pump

Function. Provide intermediate crude inventory and raise product pressure to the pipeline/export specification.

Working principle. The tank acts as a holdup node; the centrifugal pump increases liquid pressure using shaft work. The full dynamic inventory and controller behavior are not assessed in the steady-state model.



Required input / reference	Simulated stream(s) and output(s)
Feed(s)	Pump Feed Export: 554,995 kg/h at 100 kPa.
Products / connected outlet(s)	Export Crude: 554,995 kg/h at 1400 kPa; modeled Tank VENT: 0 kg/h.
Representative operating conditions	Export crude approximately 113.0°C, 1,400 kPa; Export Pump Power ≈365.5 kW.

$$\dot{W}pump \approx Q_{vol}\frac{(P_{out} - P_{in})}{\eta_{pump}} \quad (5.10)$$

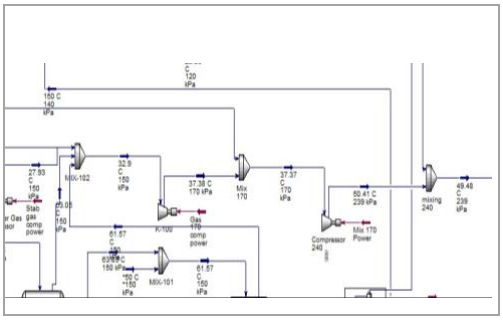
For the operating point, HYSYS explicitly provides the export pump energy stream (1.316×10⁶ kJ/h).

Engineering caution. Tank residence time, venting under upset and pump NPSH margin require equipment hydraulics/dynamics beyond this steady-state scope.

5.11 Low-Pressure Gas Recovery and Gathering

Function. Collect low-pressure vapor streams and bring them into a common higher-pressure gas handling system.

Working principle. Gas from LPDT, LPPT and the stabilizer is joined through a sequence of mixing and compression steps at approximately 150, 170, 239 and 446 kPa. Mixers conserve mass and enthalpy; compressors supply the pressure increase. Depending on source pressure, a compressor is required before gas can join a higher-pressure header.



Required input / reference	Simulated stream(s) and output(s)
Feed(s)	Stabilizer gas: 10,189; LP GAS: 935; LPDT GAS: 425 kg/h, plus dehydration/desalter gas where connected.
Products / connected outlet(s)	Gas 446 Comp feed: 12,996; Mix Gas 446: 69,561 kg/h including HP gas.
Representative operating conditions	Final pre-export-compression header gas: 69.56 t/h near 445 kPa.

$$\Sigma\,\dot{m}_{gas_{in}} = \dot{m}_{gas_{out}}\,;\quad \Sigma(\dot{m}\,h)_{in} + \dot{W}_{comp} \approx (\dot{m}\,h)_{out} \quad (5.11)$$

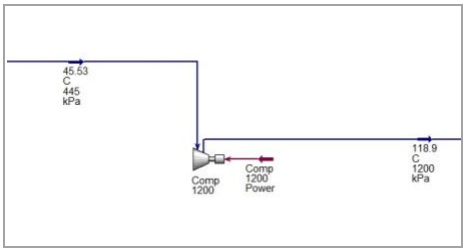
The four low-pressure gas-recovery drive energy streams are quantified in Chapter 6 and the Appendix.

Engineering caution. A gas-mixing junction cannot passively raise a lower-pressure gas to the pressure of a higher-pressure header.

5.12 Gas Compressor to 1,200 kPa

Function. Increase consolidated gas from the ~446 kPa header to the first high-pressure export-gas stage.

Working principle. The compressor performs work on a predominantly vapor stream, raising both pressure and temperature. Thermal management downstream prevents excessive temperature at the next compression stage.



Required input / reference	Simulated stream(s) and output(s)
Feed(s)	Mix Gas 446: 69,561 kg/h near 445 kPa.
Products / connected outlet(s)	Gas 1200: 69,561 kg/h at 1200 kPa.
Representative operating conditions	Gas 1200 outlet temperature 118.9°C; Comp 1200 Power ≈2,784.4 kW.

$$\dot{W}_{comp} = \dot{n}\,(h_{out} - h_{in}) \qquad (5.12)$$

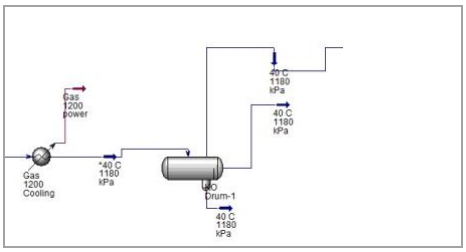
"Comp 1200 Power" = 10.024×10^6 kJ/h; motor electrical demand requires checking assumed transmission and motor efficiencies.

Engineering caution. Surge margin and compressor maps are not provided.

5.13 First Gas Cooler and KO Drum 1

Function. Cool the 1,200 kPa gas and remove condensed water and hydrocarbon liquid before the next compression step.

Working principle. Cooling releases sensible heat and can cause hydrocarbon and water condensation. The downstream knockout vessel provides phase separation so that free liquids do not enter the next compressor stage.



Required input / reference	Simulated stream(s) and output(s)
Feed(s)	Gas 1200: 69,561 kg/h; cooled feed "Gass 1200 Cooler" at 40°C.
Products / connected outlet(s)	KO GAS -1: 67,715; KO Condensate -1: 1,339; KO Water -1: 507 kg/h.
Representative operating conditions	Cooling from 118.9°C to 40°C, pressure from 1,200 to 1,180 kPa; cooler duty ≈3.653 MW.

$$\dot{m}_{cooler_{feed}} \approx \dot{m}_{KO_{vapor}} + \dot{m}_{KO_{condensate}} + \dot{m}_{KO_{water}} \qquad (5.13)$$

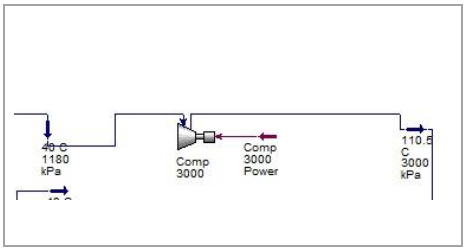
First KO condensed liquids: about 1.34 t/h hydrocarbons and 0.51 t/h water.

Engineering caution. The knockout drum phase split is theoretical; liquid carryover risk requires inlet-device, level-control and mist-eliminator design.

5.14 Gas Compressor to 3,000 kPa

Function. Raise first-KO dry-enough gas to the modeled export-compression discharge pressure.

Working principle. The second compressor receives vapor from the first knockout stage and raises its pressure from approximately 1,180 to 3,000 kPa. Compression increases temperature again, motivating a second cooler and final KO drum.



Required input / reference	Simulated stream(s) and output(s)
Feed(s)	KO GAS -1: 67,715 kg/h.
Products / connected outlet(s)	Comp 3000 outlet: 67,715 kg/h at 3000 kPa.
Representative operating conditions	Outlet T ≈110.5°C; Comp 3000 Power ≈2,428.0 kW.

$$\dot{W}_{comp_2} = \dot{n}\,(h_{3000} - h_{KO1_{gas}}) \qquad (5.14)$$

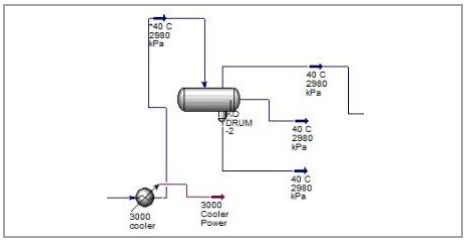
Energy stream "Comp 3000 Power" = 8.741×10^6 kJ/h.

Engineering caution. The nominal pressure ratio is approximately 2.54; final selection requires vendor surge/choke maps.

5.15 Second Gas Cooler and KO Drum 2

Function. Cool second-stage compression discharge and remove additional liquids to produce the final separated gas stream.

Working principle. The cooler reduces the 3,000 kPa discharge temperature from about 110.5 to 40°C. A final equilibrium KO vessel separates remaining hydrocarbon condensate and water from the gas.



Required input / reference	Simulated stream(s) and output(s)
Feed(s)	Comp 3000 / 3000 Cooled: 67,715 kg/h.
Products / connected outlet(s)	Gas Export - KO2: 61,500; Condensate KO-2: 6,034; Water KO-2: 181 kg/h.
Representative operating conditions	Final gas: 2,980 kPa, 40°C; second gas cooler duty ≈3.520 MW.

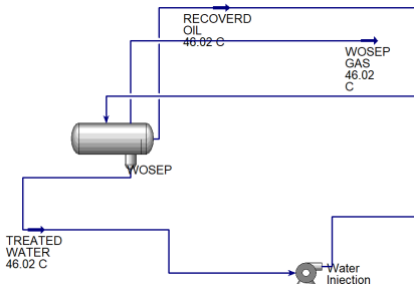
$$\dot{m}_{second\ KO\ inlet} \approx \dot{m}_{export_{gas}} + \dot{m}_{condensate} + \dot{m}_{water} \qquad (5.15)$$

The second KO removes ~6.03 t/h liquid hydrocarbon and ~0.18 t/h water.

5.16 Water–Oil Separator (WOSEP)

Function. Receive the produced-water and brine handling system and separate any free oil or gas prior to water pumping/injection.

Working principle. The upstream water streams are combined at about 150 kPa; an equilibrium three-phase vessel routes aqueous material to treated water and would route separated oil/gas to their recovery circuits when present. HYSYS currently predicts zero oil and zero gas product at WOSEP.



WOSEP -supplied HYSYS image

Required input / reference	Simulated stream(s) and output(s)
Feed(s)	Mix 150: 234,538 kg/h at 150.0 kPa; 46.0 °C.
Products / connected outlet(s)	TREATED WATER: 234,538; RECOVERD OIL: 0; WOSEP GAS: 0 kg/h.
Representative operating conditions	WOSEP outlet 46.0°C / 150 kPa; treated water flow ~234.54 t/h.

$$\dot{m}_{WOSEP\,feed} = \dot{m}_{treated\,water} + \dot{m}_{recovered\,oil} + \dot{m}_{WOSEP\,gas} \qquad (5.16)$$

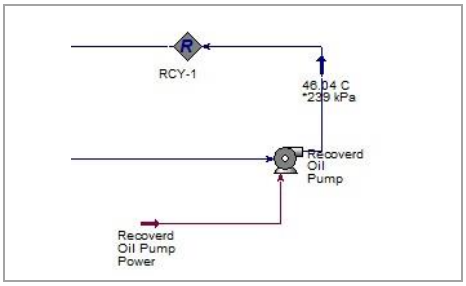
Because modeled recovered oil flow is zero, the recycle contributes no material at the reference operating point.

Engineering caution. "TREATED WATER" names a model stream; oil-in-water, solids, salinity and injection compatibility remain unverified.

5.17 Recovered-Oil Pump and Recycle

Function. Return separated oil from WOSEP to upstream liquid processing, recovering hydrocarbons that would otherwise accompany water handling.

Working principle. The return liquid is pressurized and recycled to a suitable inlet point. A HYSYS recycle convergence block mathematically closes the loop, but the entire system can converge at zero return-oil flow.



Required input / reference	Simulated stream(s) and output(s)
Feed(s)	RECOVERD OIL: 0 kg/h.
Products / connected outlet(s)	Recoverd Oil recycle: 0 kg/h; LPPT+ Recvored: 568,700 kg/h including primary feed.
Representative operating conditions	Recovered Oil Pump Power = 0 kW in the supplied energy-stream data.

$$\dot{m}_{recycle} = \dot{m}_{recovered\ oil} - \dot{m}_{losses} \qquad (5.17)$$

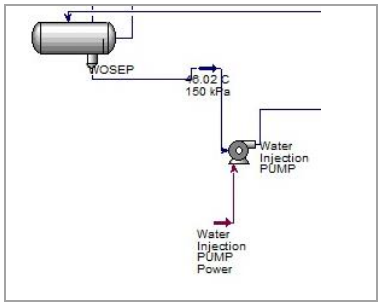
The recovery loop should be analyzed again if nonzero recovered oil is produced.

Engineering caution. A reported 0 kW pump load is not evidence that a physical recycle pump consumes no energy at nonzero flow.

5.18 Produced-Water Injection Pump

Function. Raise the water stream pressure sufficiently for a modeled downstream injection/disposal system.

Working principle. The pump takes the WOSEP aqueous outlet and supplies hydraulic energy. Unlike a separator, it does not remove dissolved salts, H2S, hydrocarbons or particulates; those require additional treatment specification if injection is a real project objective.



Required input / reference	Simulated stream(s) and output(s)
Feed(s)	TREATED WATER: 234,538 kg/h at 150 kPa.
Products / connected outlet(s)	Water Injection PUMP: 234,538 kg/h at 5000 kPa.
Representative operating conditions	Outlet pressure ~5,000 kPa; power ≈425.0 kW.

$$\dot{W}_{water\ pump} \approx \frac{Q_{water}(5{,}000 - 150)kPa}{\eta_{pump}} \qquad (5.18)$$

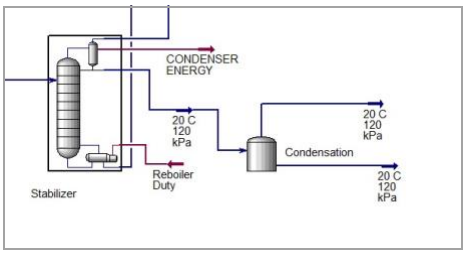
Computed energy-stream load is 1.530×10^{6} kJ/h.

Engineering caution. Actual injection acceptance also requires water chemistry, filtration, well injectivity and pipeline hydraulics.

5.19 Condensate Tank and Product Routing

Function. Collect the liquid overhead condensate produced by partial condensation in the stabilizer.

Working principle. The condensate stream is withdrawn to a product/storage route, while any tank vapor outlet is represented separately. For the reference model, the condensate vent stream is zero and condensate mass flow is about 0.76 tonnes/h.



Required input / reference	Simulated stream(s) and output(s)
Feed(s)	Stabilizer Condensate: 760.5 kg/h.
Products / connected outlet(s)	Condensate Product: 760.5 kg/h; Tank Condensate Vent: 0 kg/h.
Representative operating conditions	Overhead liquid T ≈20°C, pressure ≈120 kPa.

$$\dot{m}_{condensate\,in} = \dot{m}_{condensate\,product} + \dot{m}_{tank\,vent} \qquad (5.19)$$

Storage design should consider flash pressure, volatility and safety relief.

Engineering caution. A zero modeled vent at one operating point is not a substitute for tank breathing/flashing and relief calculations.

5.20 Stabilizer internal temperature and pressure profiles

The simulated column exhibits a strong upward-to-downward temperature gradient: the condenser is 20°C; tray 1 is 30.50°C; tray 6 rises to 71.49°C; tray 12 reaches 110.9°C; and the reboiler reaches 150°C. Pressure rises from 120 kPa at the condenser to 140 kPa at the reboiler. This trend is consistent with stronger stripping and a heavier liquid phase toward the bottom; it does not establish tray hydraulics or mechanical pressure rating.

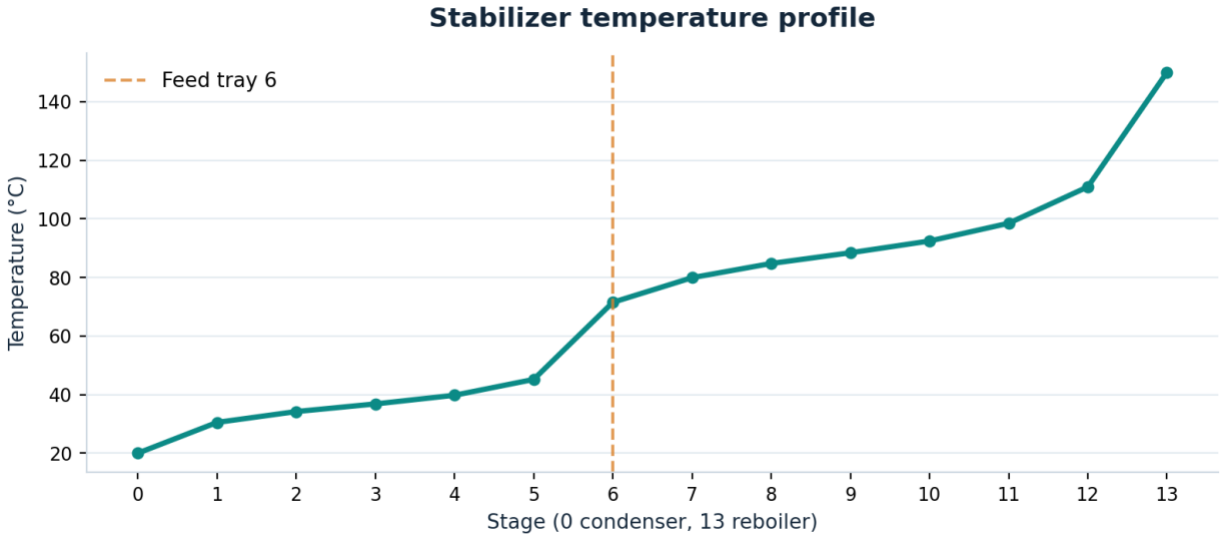


Figure 5.2. HYSYS stabilizer temperature profile (stage-by-stage).

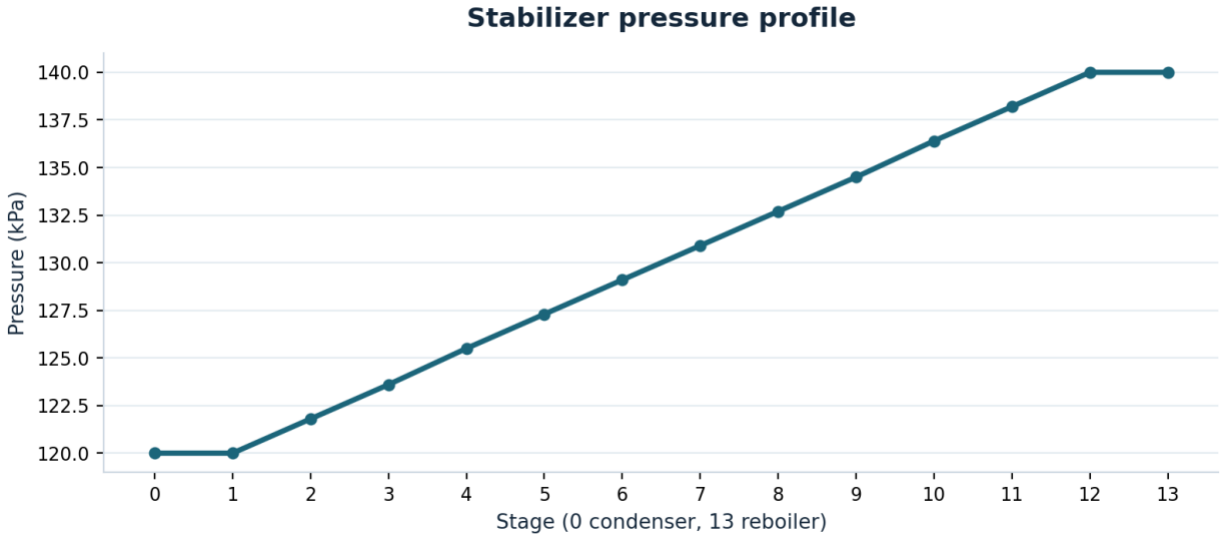


Figure 5.3. HYSYS stabilizer pressure profile (stage-by-stage).

6. MATERIAL AND ENERGY BALANCE VERIFICATION

6.1 Equipment-level checks

Three-phase separator checks were performed directly against named workbook streams. Where HYSYS reports zero separated water or oil, the mass closure is still satisfied but phase-separation performance must not be overstated. Selected calculations illustrate closure with rounded values:

$$HPPT: 792{,}658 \approx 56{,}566 + 568{,}700 + 167{,}393 \quad kg/h \qquad (6.1)$$

$$LPPT: 568{,}700 \approx 935 + 567{,}764 + 0 \quad kg/h \qquad (6.2)$$

$$KO-2: 67{,}715 \approx 61{,}500 + 6{,}034 + 181 \quad kg/h \qquad (6.3)$$

The few kg/h discrepancies in these displayed sums arise from rounding of the printed engineering values; the underlying workbook has greater numerical precision.

6.2 Overall facility system boundary

To avoid double counting, overall inlet flows comprise the three primary feed streams plus the two external desalting wash-water supplies. Overall outlet streams comprise the export crude, export gas, separated knockout condensates and knockout waters, stabilized overhead condensate, the treated/injection water stream and the small condenser aqueous product. Water drawn from Desalter 1 and Desalter 2 is internal to the modeled downstream water-treatment route, not an extra external plant outlet. Similarly, recycle flow must not be counted as fresh feed.

Using the workbook and approximately 39.2 kg/h condenser aqueous outlet from the stabilizer screenshot, calculated external input is 859,894.4 kg/h and external output is 859,894.4 kg/h. The difference is -0.0 kg/h (-0.0000% of total feed). This is a bookkeeping and conservation check, not independent experimental validation of crude product quality.

6.3 Utility load classification

The modeled shaft-demand energy streams sum to approximately 6.289 MW. The reboiler heating duty is 30.977 MW thermal. The four identifiable cooling services sum to 13.027 MW thermal. The first two quantities must not be added indiscriminately to the cooling demand as though all energy streams were purchased electricity: the cost of cooling depends on utility technology, and reboiler heat may be delivered by steam or direct-fired heating.

Energy service	Demand	Definition
Pumps and compressors	6.289 MW shaft	Sum of eight named rotating-equipment energy streams
Stabilizer reboiler	30.977 MW thermal	Positive reboiler energy stream
Four modeled coolers	13.027 MW thermal	Condenser + crude + two gas coolers

7. SENSITIVITY STUDIES (CASE STUDIES)

7.1 Study 1: linked inlet throughput variation (80–120%)

The flow of Arabian Light Crude, ASSOCIATED GAS and PRODUCED WATER was varied simultaneously by common multipliers of 0.80, 0.90, 1.00, 1.10 and 1.20. This preserves the initial feed proportions in the specified streams. The exported crude, exported gas and reboiler duty increase approximately in proportion to total feed. The 100% reference case gives 555.0 t/h export crude and 61.49 t/h export gas; the 120% case gives 666.0 and 73.79 t/h, respectively. HYSYS converged all five cases, but this trend does not show whether vessel residence time, equipment capacity and compressor surge margins remain acceptable at higher throughput.

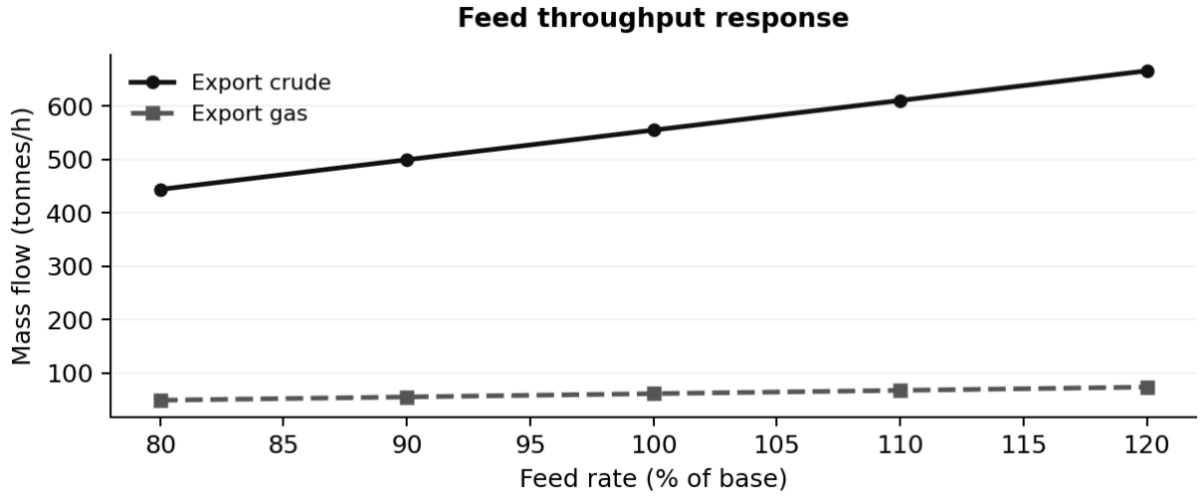


Figure 7.1. Exported product throughput under linked fresh-feed flow changes.

In the originally exported case-study worksheet, “Crude Cooler – Heat Flow” is a material stream enthalpy flow (approximately −1.089×10⁹ kJ/h at the base case), not the duty of the physical crude cooler. The actual cooler-duty estimate is taken from the Energy Streams worksheet, preventing a major apparent heat-duty overstatement.

7.2 Study 2: stabilizer reflux ratio and volatility

Reflux ratio was varied from 0.40 to 0.60, while the other active stabilizer specifications remained as recorded in HYSYS. A larger ratio enhances internal liquid recycle and changes the distribution of overhead liquid and vapor. The stabilizer gas rate rises from approximately 9.80 to 10.45 t/h, while overhead condensate falls from 1.277 to 0.415 t/h. The model-calculated Reid VP of the stabilized crude falls from 6.267 to 6.240 kPa, a reduction of just 0.027 kPa (about 0.43%). Meanwhile condenser energy-stream duty increases from 2.808 to 3.420 million kJ/h, roughly 21.8%.

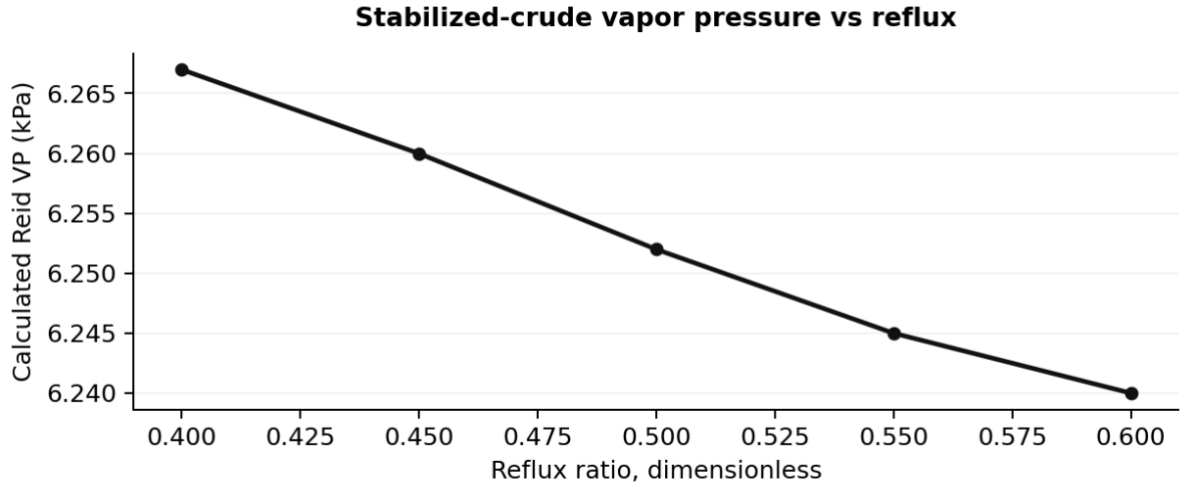


Figure 7.2. Marginal model-calculated RVP improvement as reflux increases.

This is a clear diminishing-returns result for the measured RVP quantity, not proof that the base case reflux ratio is a global economic optimum. A justified optimum would require a target RVP, a crude value/condensate value basis, operating constraints, possible stage efficiencies and price-based marginal utility comparisons.

7.3 Study 3: HPPT inlet pressure

An important modeling distinction arose during this study. Directly selecting the HYSYS variable "HPPT – Vessel Pressure" did not change the result; every trial still reported 446 kPa. The operative case-study input was instead "HPPT FEED – Pressure." Therefore the evidenced experiment must be described as an HPPT feed-pressure study, not automatically a controlled separator-vessel-pressure optimization. HYSYS inlet-pressure trial points were 446, 470, 500, 530 and 560 kPa; the HP GAS output decreases from about 56.57 to 54.17 t/h across the states, while crude and final export gas change only slightly.

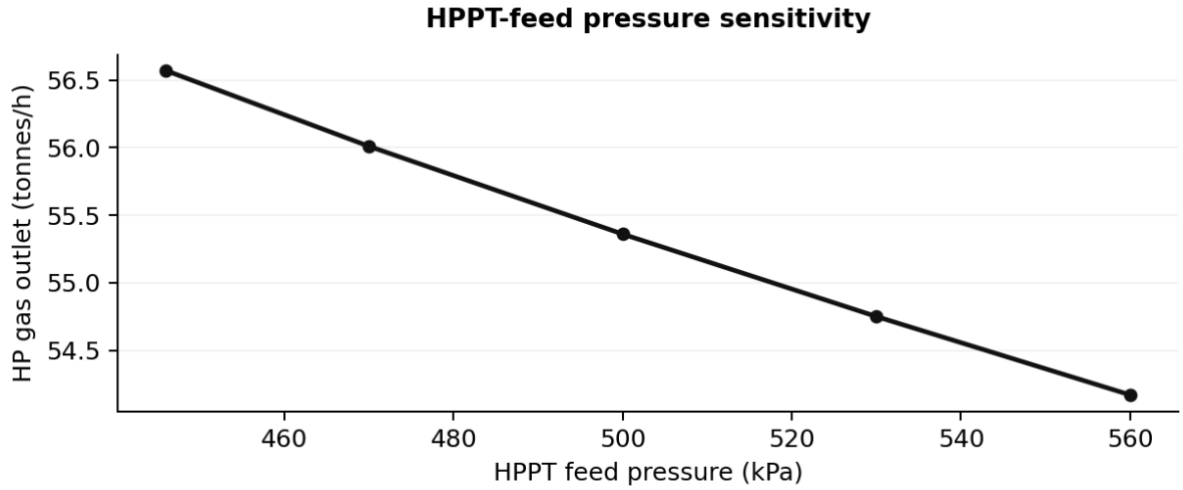


Figure 7.3. Simulated HP gas mass flow as the HPPT feed-pressure specification changes.

Before any physical mechanism is assigned unequivocally to these results, the actual HPPT vessel and outlet pressures should be logged at every state to confirm that the separator pressure, rather than only the upstream

boundary or an upstream flash, changed. The observed mass-flow trend is a valid HYSYS response, but the causal interpretation carries this verification requirement.

7.4 Concise numeric summary of case results

Case study	Base condition	Range tested	Key observation
Throughput	100% of feed	80–120%	Export crude: 444 to 666 t/h
Reflux	R = 0.50	0.40–0.60	RVP: 6.267 to 6.240 kPa
HPPT feed pressure	446 kPa	446–560 kPa	HP gas: 56.57 to 54.17 t/h

The three studies demonstrate that different operating parameters affect GOSP performance in different ways.

Feed throughput has the strongest direct influence on production rates and total processing energy requirements. Stabilizer reflux ratio primarily affects condenser duty, condensate recovery, and crude volatility. HPPT feed pressure influences the initial gas separation behavior but has a relatively limited effect on final production.

Overall, the results highlight the importance of balancing production, product quality, and energy consumption when selecting operating conditions. Steady-state sensitivity analysis provides a useful foundation for process optimization and identifying promising operating ranges; however, equipment constraints, product specifications, and economic trade-offs must be considered before determining an optimal operating point.

8. PRELIMINARY ECONOMIC ASSESSMENT

8.1 Scope and assumptions

A simple utility operating-cost screening is presented for comparison of energy demands, not as a full GOSP operating budget. To avoid attributing the result to a particular utility contract, all prices and operating assumptions are explicit: 8,000 operating h/year, electricity at 0.20 SAR/kWh, heating supplied with an assumed 85% efficient fuel/steam system and a reference fuel price of 8.06 SAR/MMBtu. An illustrative cooling circulation equivalent is calculated for 10°C water temperature rise and valued at 69.16 SAR/1,000 m³. These are scenario inputs, not independently verified procurement prices for the facility.

8.2 Calculation

$$Celectric = Pshaft\ (kW) \times 8{,}000\ (h/y) \times 0.20\ (SAR/kWh) \quad (8.1)$$

$$C_{fuel} = [\frac{Q_{reboiler}\left(\frac{GJ}{h}\right)}{0.85}] \div 1.05506\ (GJ/MMBtu) \times 8{,}000 \times 8.06 \quad (8.2)$$

$$\dot{V}_{cool\ water} \approx \frac{\dot{Q}_{cool}}{\rho\ Cp\ \Delta T} \quad (8.3)$$

With the stated screening basis, modeled electricity cost is about **10.06 million SAR/year**, heating cost about **8.02 million SAR/year**, and cooling circulation charge about **0.62 million SAR/year**. The sum is approximately **18.70 million SAR/year**. Depending on the actual utility arrangement, the circulating water might not be billed by fresh makeup volume; thus this third line is particularly technology- and contract-dependent.

Utility expense	Illustrative SAR million/year
Rotating-equipment electricity	10.06
Reboiler heating fuel	8.02
Cooling-water circulation equivalent	0.62
Total modeled utility screening cost	18.70

The model excludes revenue, actual steam generation scheme, water treatment chemicals, maintenance, staffing, corrosion inhibitor, capital charges, equipment replacement and offsite charges. Equipment utility efficiency should be verified with design datasheets before comparison to an operating plant.

9. CORROSION ASSESSMENT

Steady-State Prediction for the HPPT Feed and WOSEP Feed Pipelines This standalone section evaluates the predicted uniform corrosion of an assumed carbon-steel piping material at two selected locations in the Gas Oil Separation Plant (GOSP): the multiphase HPPT inlet and the water-rich stream entering the Water–Oil Separator (WOSEP). Aspen HYSYS V14 supplies fluid properties and flow rates; FREECORP 2.2 predicts bare-steel electrochemical corrosion under specified water chemistry.

9.1 Theoretical background

Internal corrosion requires an electrochemically active, electrically conductive phase in contact with the metal. Dissolved CO_2 can form carbonic acid; dissolved H_2S affects sulfide chemistry, electrochemical reactions and possible iron-sulfide films. Carbon-steel dissolution is represented by the anodic half-reaction:

$$Fe \rightarrow Fe^{2+} + 2e^{-}$$

Cathodic processes may include reduction of hydrogen ions and water, as well as reactions involving dissolved acid gases. The intersection of the net anodic and cathodic polarization curves on an Evans diagram establishes the modelled corrosion potential and current density. By Faraday's law, the uniform penetration rate is proportional to the corrosion current density:

$$CR = \left(\frac{i_{corrM}}{nF\rho}\right) \times (seconds\ per\ year) \times 1000 \quad [mm/year]$$

where i_corr is corrosion current density (A/m²), M is metal molar mass (kg/mol), n is the electron number, F is Faraday's constant (C/mol), and ρ is metal density (kg/m³). FREECORP resolves the electrochemical reactions numerically; this equation explains the conversion to a material-loss rate rather than serving as an independent re-calculation.

9.2 Workflow and assumptions

For each stream, HYSYS phase data were transferred to FREECORP with the selected simulation options: Point geometry, BLC (Free), Water Flow and Steady State. The superficial water velocity was estimated from aqueous mass flow, water density and the assumed internal pipe diameter:

$$v_sw = \frac{\left(\frac{\dot{m}_{water}}{\rho_{water}}\right)}{\left[3600 \times \frac{\pi D^2}{4}\right]}$$

Input category	Treatment in this study
From HYSYS	Temperature, pressure, phase split, aqueous mass flow, selected compositions and density
Assumed / scenario	Carbon steel; internal diameter 0.570 m; pH values of 4, 5 and 6; NaCl 1 wt%; Fe^{2+} 1 ppm; dissolved O_2 0 ppm
Model limitation	No field pH / salinity, metallurgy, corrosion-monitor data or model calibration; results are screening predictions

9.3 Scope of interpretation

The reported rates refer to predicted uniform attack of unprotected mild steel under the chosen water-wetting and chemistry assumptions. They do not quantify pitting, erosion-corrosion, sulfide stress cracking, hydrogen-induced cracking or actual remaining pipe life. The low CO_2 concentration triggered a FREECORP soft-limit warning; this is an additional reason to interpret the values as exploratory rather than validated design rates. Original HYSYS input records:

Stream Name	HPPT FEED	Vapour Phase	Liquid Phase	Aqueous Phase
Vapour / Phase Fraction	0.1698	0.1698	0.1828	0.6474
Temperature [C]	39.41	39.41	39.41	39.41
Pressure [kPa]	446.0	446.0	446.0	446.0
Molar Flow [kgmole/h]	1.435e+004	2437	2623	9290
Mass Flow [kg/h]	7.927e+005	5.657e+004	5.687e+005	1.674e+005
Std Ideal Liq Vol Flow [m3/h]	982.3	147.0	667.6	167.7
Molar Enthalpy [kJ/kgmole]	-2.834e+005	-8.697e+004	-4.602e+005	-2.851e+005
Molar Entropy [kJ/kgmole-C]	152.8	181.0	464.4	57.40
Heat Flow [kJ/h]	-4.067e+009	-2.120e+008	-1.207e+009	-2.648e+009
Liq Vol Flow @Std Cond [m3/h]	920.6	5.735e+004	665.5	165.0
Fluid Package	Basis-2			

Stream Name	Mix 150	Aqueous Phase
Vapour / Phase Fraction	0.0000	1.0000
Temperature [C]	46.02	46.02
Pressure [kPa]	150.0	150.0
Molar Flow [kgmole/h]	1.302e+004	1.302e+004
Mass Flow [kg/h]	2.345e+005	2.345e+005
Std Ideal Liq Vol Flow [m3/h]	235.0	235.0
Molar Enthalpy [kJ/kgmole]	-2.845e+005	-2.845e+005
Molar Entropy [kJ/kgmole-C]	59.03	59.03
Heat Flow [kJ/h]	-3.704e+009	-3.704e+009
Liq Vol Flow @Std Cond [m3/h]	231.1	231.1
Fluid Package	Basis-2	

Figure 9.1. Original HYSYS stream conditions: multiphase HPPT feed (left) and water-rich Mix 150 / WOSEP feed (right).

9.4 HPPT feed pipeline

The HPPT inlet transports a mixed oil–gas–water stream before high-pressure phase separation. HYSYS predicts a free aqueous phase, so a hypothetical water-wetted bottom-of-line carbon-steel surface is a relevant screening case. The aqueous mass flow and assumed diameter give a superficial water velocity of approximately 0.184 m/s.

Process / model input	Value	Basis
Temperature; total pressure	39.41 °C; 4.46 bar	HYSYS (bar assumed absolute)
Free-water mass flow	167,400 kg/h	HYSYS aqueous phase
CO_2; H_2S in vapour	1.919%; 3.491 mol%	HYSYS phase composition
Pipe inner diameter; v_sw	0.570 m; 0.184 m/s	Assumed diameter; calculated velocity
Aqueous chemistry	pH 4/5/6; 1 wt% NaCl	Scenario assumptions

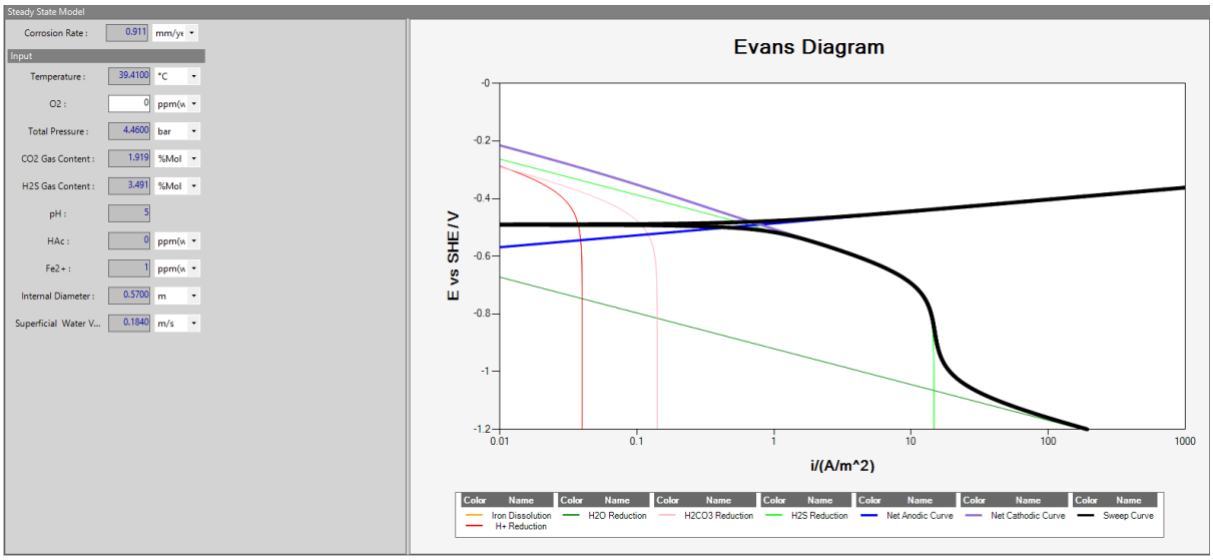


Figure 9.2. Original FREECORP result for the HPPT feed at pH 5 (predicted corrosion rate: 0.911 mm/year).

Assumed pH	4.0	5.0 (base)	6.0
Predicted CR (mm/year)	1.220	0.911	0.881

The predicted rate fell by 25.3% when pH increased from 4 to 5 and by a further 3.3% from pH 5 to 6. At lower pH, the model predicts stronger acid-driven cathodic activity and a higher uniform corrosion current. The response is non-linear: raising pH from 5 to 6 yields a smaller additional reduction than raising it from 4 to 5.

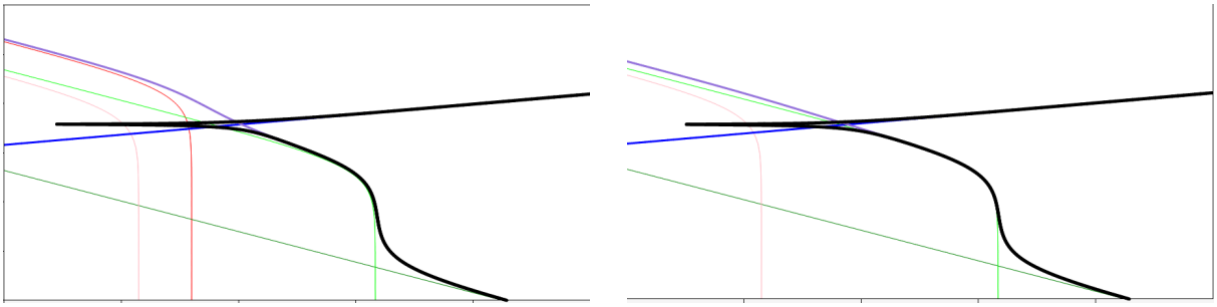


Figure 9.3. User-supplied Evans-diagram excerpts at pH 4 (left) and pH 6 (right), HPPT feed.

9.5 WOSEP feed pipeline (Mix 150)

The Mix 150 stream enters WOSEP and is predicted to be entirely aqueous at the HYSYS operating point. Water wetting is therefore a reasonable screening basis. Unlike the HPPT feed, however, this stream has no modelled free vapour phase: equivalent acid-gas partial pressures had to be imposed as an explicit scenario for FREECORP rather than transferred directly from a vapour composition.

Process / model input	Value	Basis
Temperature; total pressure	46.02 °C; 1.50 bar	HYSYS (bar assumed absolute)
Aqueous mass flow; density	234,500 kg/h; 991.309 kg/m³	HYSYS
CO_2 gas-equivalent pressure	0.010 bar	Assumed illustrative value
H_2S gas-equivalent pressure	0.200 bar	Preliminary equilibrium estimate; unvalidated
Pipe inner diameter; v_sw	0.570 m; 0.258 m/s	Assumed diameter; calculated velocity
Aqueous chemistry	pH 4/5/6; 1 wt% NaCl	Scenario assumptions

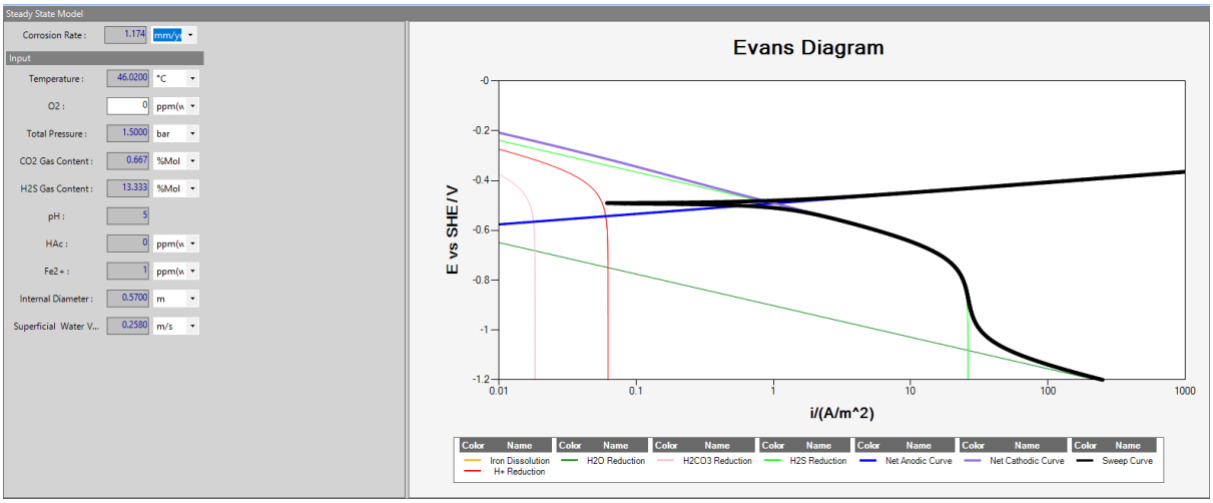


Figure 9.4. Original FREECORP result for the WOSEP feed at pH 5 (predicted rate: 1.174 mm/year).

Assumed pH	4.0	5.0 (base)	6.0
Predicted CR (mm/year)	1.637	1.174	1.129

From pH 4 to 5, the rate decreased by 28.3%, whereas the reduction from pH 5 to 6 was only 3.8%. The simulated acidic scenario therefore gives the greatest modelled corrosion. These results are particularly sensitive to the imposed CO_2/H_2S equilibrium pressures, which are not measured stream conditions; a real-water sample and suitable equilibrium characterization would be needed for validation.

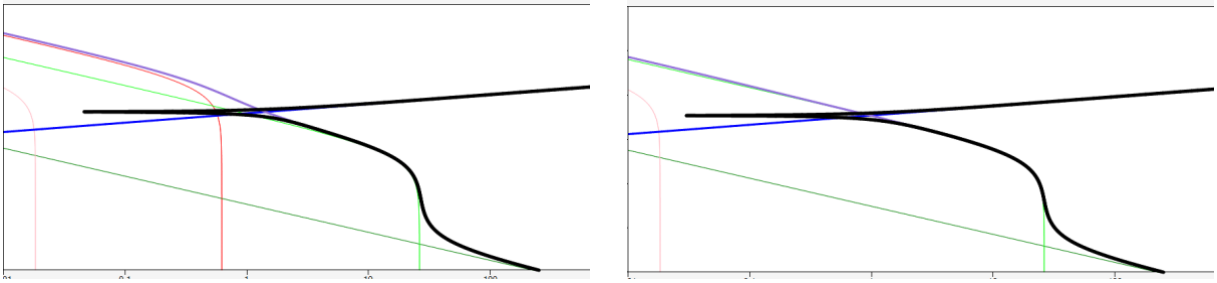


Figure 9.5. User-supplied Evans-diagram excerpts at pH 4 (left) and pH 6 (right), WOSEP feed.

9.6 Comparative discussion and conclusions

The two independent FREECORP scenarios show the same directional response: increasing assumed water pH decreases the predicted uniform corrosion rate. The difference is largest over pH 4–5 and becomes much smaller over pH 5–6. The WOSEP scenario produces the higher numerical prediction at each investigated pH, but this comparison is conditional on different acid-gas input quality, flow conditions and an assumed common steel grade.

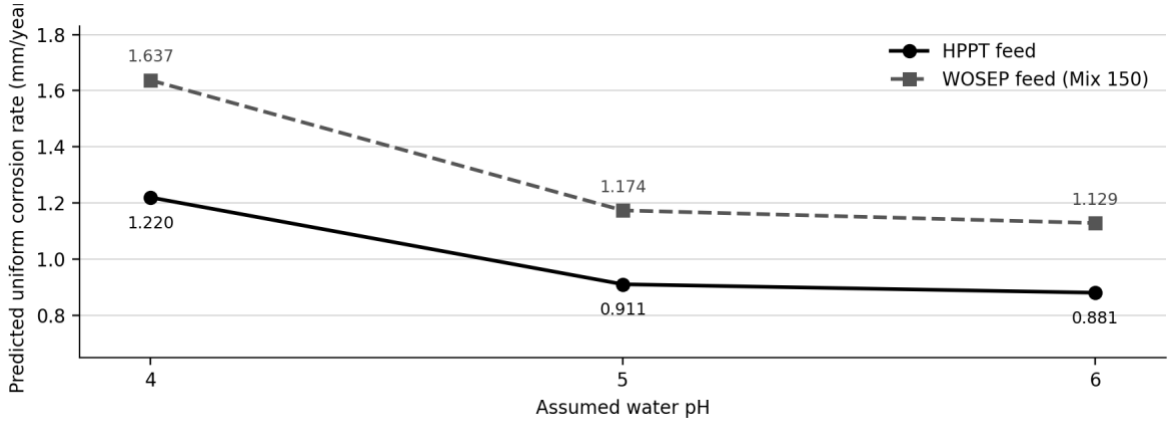


Figure 9.6. Corrosion-rate sensitivity to assumed pH, reconstructed from the six user-provided FREECORP results.

Stream	pH 4	pH 5 (base)	pH 6	pH 4 → 6
HPPT feed	1.220	0.911	0.881	−27.8%
WOSEP feed	1.637	1.174	1.129	−31.0%

9.6.1 Engineering interpretation

At the reference pH of 5, the WOSEP prediction exceeds the HPPT feed prediction by 0.263 mm/year (28.9%). This difference must not be attributed solely to temperature, water velocity or hydrogen sulfide: several conditions differ simultaneously. In particular, WOSEP CO_2 and H_2S gas-equivalent inputs were assumed or estimated for an aqueous-only stream, while HPPT acid-gas inputs came from a modelled vapour composition.

9.6.2 Reliability and recommended next work

For an academic sensitivity study, both datasets demonstrate reproducible numerical trends from FREECORP 2.2. For materials selection or field integrity decisions, neither set is yet validated: pH, ionic content, pipe dimensions and carbon-steel specification should be confirmed; dissolved acid gases should be measured or estimated with a validated aqueous-equilibrium model; software input-range warnings should be resolved; and corrosion coupons/probes or inspection data should be used to test predicted rates. Sour-service cracking and localized corrosion require separate assessments.

9.6.3 Summary statement

Within the selected model scenarios, lower pH corresponds to higher predicted uniform carbon-steel corrosion. Reference rates at pH 5 were 0.911 mm/year (HPPT feed) and 1.174 mm/year (WOSEP feed). These values are model-based, assumption-sensitive estimates—not measured plant corrosion rates.

Data attribution: All operational inputs and Evans diagrams originate from the user's Aspen HYSYS V14 and FREECORP 2.2 screenshots (October 2026). The comparison chart is based on those six simulation outputs. Modeling basis: Ohio University, Institute for Corrosion and Multiphase Technology, FREECORP 2.2; AspenTech, Aspen HYSYS V14.

10. TECHNICAL LIMITATIONS AND RECOMMENDED WORK

10.1 Model-validation priorities

First, verify the pseudo-component and light-end composition against reliable crude PVT/assay data, because the resulting VLE determines flash-gas rates, stabilizer duty and Reid VP. Second, confirm all actual pressures, especially the HPPT feed-pressure case-study states. Third, supplement the simplified desalting model with realistic water-in-crude, chloride, salt-transfer and separation-efficiency specifications. Fourth, verify compressor maps and product-gas hydrocarbon/water dew-point limits; a knockout stream alone does not certify dryness or commercial pipeline quality.

10.2 Mechanical and operating checks

Separator gravity capacity and retention times, level-control philosophy, compressor surge limits, gas cooler fouling, crude exchanger design, pump NPSH and control-valve sizing should be evaluated in later design stages. Product-gas H2S and water content should be compared to a stated sales-gas specification if such a claim becomes an objective. No such acceptance criteria are supplied here.

10.3 Scope of future dynamic work

Preliminary dynamic conversion work was attempted, including vessel holdup sizing, valve pressure–flow relationships and level-controller connections. Plant-wide transient convergence and controller tuning were not completed or validated. Consequently, no step-response times, shutdown performance, trip timings, controller stability or process-upset recovery claims are included in this steady-state academic report.

11. CONCLUSION

The developed HYSYS model provides an integrated steady-state representation of an oilfield wellstream conditioning system. Three-stage pressure separation performs the initial oil–gas–water split, followed by crude heating, phase-based dehydration, two wash-water separation stages and a staged crude stabilizer. The gas route uses successive compression, cooling and knockouts before final export; aqueous streams enter water handling and injection. Equipment descriptions in this study explicitly connect physical purpose to stream data and governing equations, rather than treating unit operations as independent schematic symbols.

At the reported baseline the facility exports approximately 555.0 t/h crude and 61.5 t/h gas, while the stabilizer demands about 31.0 MW of thermal input and the modeled rotating equipment about 6.29 MW of shaft power. Reflux sensitivity demonstrates small model-calculated RVP benefit for substantially greater condenser duty over the tested range. The throughput study demonstrates nearly proportional production and duty changes; the HPPT inlet pressure study produces measurable flow redistribution that still requires confirmation of the actual separator pressure response. The recommended next steps are assay/PVT validation, equipment and water-quality verification, better constrained economic optimization and the separately reserved corrosion chapter.

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APPENDIX A. SELECTED REFERENCE STREAM DATA

Selected material stream	Mass flow (kg/h)	T (°C)	P (kPa)
HPPT FEED	792,658.4	39.41	446.0
HP GAS	56,565.6	39.41	446.0
HP OIL	568,699.5	39.41	446.0
HP WATER	167,393.2	39.41	446.0
LPPT FEED	568,699.5	39.26	239.0
LP GAS	935.1	39.26	239.0
LPDT GAS	424.9	39.18	170.0
DEHYD CRUDE	566,180.4	65.00	150.0
DESALT CRUDE-1	565,996.0	63.05	150.0
Desalt CRUDE-2	565,983.7	61.57	150.0
Stabilizer Crude	554,995.4	150.00	140.0
Stabilizer gas	10,188.6	20.00	120.0
Stabilizer Condensate	760.5	20.00	120.0
Export Crude	554,995.4	112.97	1,400.0
Mix Gas 446	69,561.2	45.52	445.0
Gas 1200	69,561.2	118.90	1,200.0
KO GAS -1	67,715.3	40.00	1,180.0
Comp 3000	67,715.3	110.51	3,000.0
Gas Export - KO2	61,500.2	40.00	2,980.0
Mix 150	234,538.1	46.02	150.0
TREATED WATER	234,538.1	46.02	150.0

This appendix intentionally lists only streams necessary to trace the process and validate unit balances. The complete Material Streams workbook remains the controlling source for all material-stream fields.

APPENDIX B. SELECTED FEED AND PRODUCT COMPONENT DATA

Mole fraction	Arabian Light Crude	ASSOCIATED GAS	Stabilizer Crude	Gas Export - KO2
H2O	0	0	9.245e-15	0.002788
CO2	0	0.02	4.669e-16	0.01923
H2S	0	0.04	4.087e-12	0.03722
Methane	0	0.72	3.394e-18	0.7015
Ethane	0.0004936	0.09	2.484e-13	0.08646
Propane	0.004799	0.06	1.396e-09	0.05908
n-Butane	0.01418	0.025	3.729e-06	0.03137

Heavy-oil pseudo-components are omitted from this compact table to retain the theoretical emphasis of the main report. Their full HYSYS composition is present in the submitted Excel workbook.

APPENDIX C. THREE DISCRETE CASE STUDIES

C.1 Feed throughput (selected calculated outputs)

Feed (%)	Crude (kg/h)	Gas (kg/h)	Reboiler (10^6 kJ/h)
80	444,000	49,200	89.78
90	499,500	55,340	100.70
100	555,000	61,490	111.50
110	610,500	67,640	122.40
120	666,000	73,790	133.20

C.2 Reflux ratio (selected calculated outputs)

Reflux R	RVP (kPa)	Condenser (10^6 kJ/h)	Condensate (kg/h)
0.40	6.267	2.808	1277.0
0.45	6.260	2.932	999.6
0.50	6.252	3.076	765.4
0.55	6.245	3.240	573.3
0.60	6.240	3.420	415.1

C.3 HPPT feed-pressure sensitivity (selected calculated outputs)

Feed P (kPa)	HP GAS (kg/h)	Export crude (kg/h)	Export gas (kg/h)
446	56,570	555,000	61,500
470	56,010	555,100	61,520
500	55,360	555,100	61,550
530	54,750	555,200	61,570
560	54,170	555,300	61,600

APPENDIX D. SOURCE FIGURES FOR TRACEABILITY

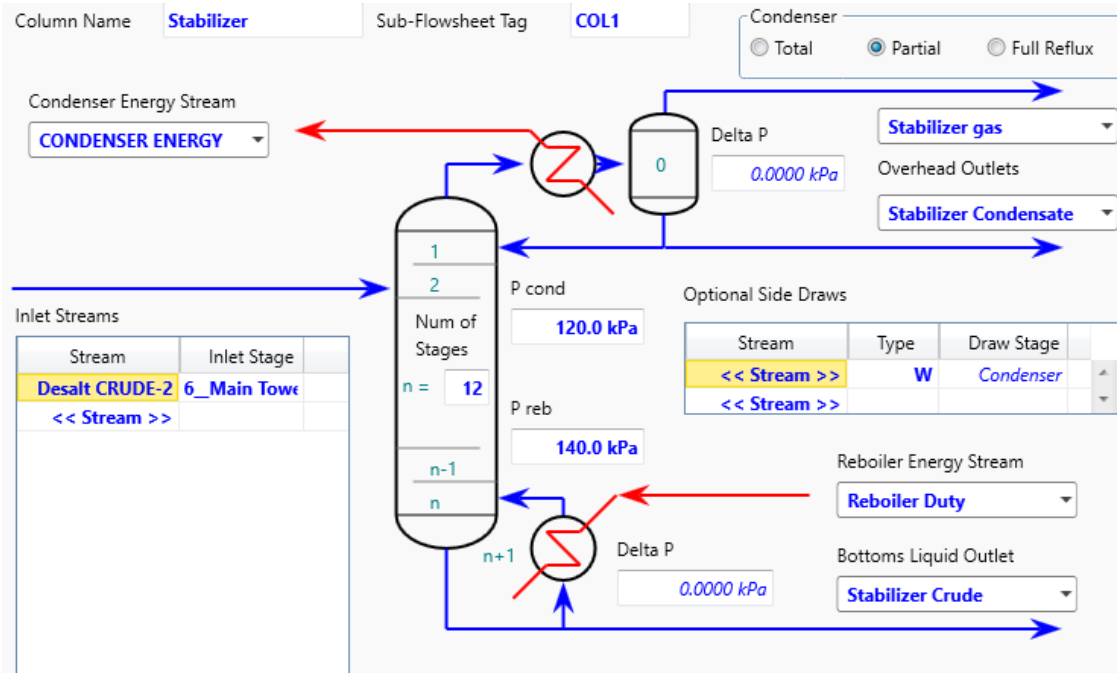


Figure D.1. User-supplied HYSYS stabilizer configuration and feed/condenser/reboiler settings.

Name	Desalt CRUDE-2 @COL1	Stabilizer gas @COL1	Stabilizer Condensate @COL1	Condenser_au... @COL1	Stabilizer Crude @COL1
Vapour	0.0000	1.0000	0.0000	0.0000	0.0000
Temperature [C]	61.57	20.00	20.00	20.00	150.0
Pressure [kPa]	150.0	120.0	120.0	120.0	140.0
Molar Flow [kgmole/h]	2548	177.2	10.80	2.176	2358
Mass Flow [kg/h]	5.660e+005	1.019e+004	760.5	39.20	5.550e+005
Std Ideal Liq Vol Flow [m3/h]	661.9	17.39	1.204	3.928e-002	643.2
Molar Enthalpy [kJ/kgmole]	-4.612e+005	-1.274e+005	-1.659e+005	-2.866e+005	-4.418e+005
Molar Entropy [kJ/kgmole-C]	503.3	170.9	93.93	52.40	651.5
Heat Flow [kJ/h]	-1.175e+009	-2.258e+007	-1.791e+006	-6.235e+005	-1.042e+009

Figure D.2. User-supplied HYSYS column feed and outlet properties.